Assignment – 2

Programming and Problem Solving

COMP 6481 --- Fall 2026

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Artémis Mission Control

Assignment Overview

A Java program that:

1. Manages mission operations using Array List and LinkedList structures.

2. Supports adding, updating, deleting, searching, cloning, & completing mission tasks.

3. Identifies mission-critical operations based on priority.

4. Detects mission tasks whose scheduled execution time has passed.

5. Handles invalid, incomplete, or duplicate information using custom exceptions.

6. Provides user interaction through a console-based menu.

7. Implements object cloning, copy constructors, deep copies, and object comparison.

8. Generates summaries of important and completed mission operations.

9. Demonstrates appropriate use of encapsulation, interfaces, classes, objects, and collection data structures.

Task Description

Mission Critical interface

Interface defining the isCritical() contract

Mission Task

Core data class representing a single mission operation with all attributes, constructors, clone(), equals(), and toString().

Mission Timeline

Custom linked list built from scratch using an inner node class. Supports insert, delete, replace, search, and comparison operations.

Mission Control Manager

Manages active tasks (Array List) and completed tasks (LinkedList). Contains all business logic methods.

Invalid Mission Task Exception

Custom exception thrown when mission task data is missing or invalid.

Duplicate Mission Task Exception

Custom exception thrown when a duplicate task ID is detected.

Artemis Mission Control Main

Main driver class with the console menu and user interaction logic.

Array List / LinkedList

1. Active mission tasks are stored in an Array List
2. Completed tasks are stored in a LinkedList

Deep Copy / Shallow Copy

1. The required Systems Array List inside Mission Task is deep copied in both the copy constructor and the clone() method
2. A shallow copy would share the same list reference between two objects which could cause unexpected changes.

Custom Linked List (Mission Timeline)

1. Mission Timeline is built from scratch without using Java's built-in LinkedList class.
2. It uses an inner class called MissionTaskNode where each node holds a MissionTask object and a reference to the next node.
3. If an invalid index is given in the input to insertAtIndex(), deleteFromIndex(), or replaceAtIndex(), a NoSuchElementException is thrown.

Interface Usage

1. The Mission Critical interface defines a single method isCritical() which Mission Task implements.
2. The method returns true only when the task priority is between 8 and 10 inclusive.

Exception Handling

Two custom exception classes

1) Invalid Mission Task Exception

Thrown when a task is created with missing or invalid data such as an empty task.

2) Duplicate Mission Task Exception

Thrown when a task ID that already exists in the active list is used for a new task.